# Answers by exercise and question

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## Exercise 1 · Vector products, angles, areas and volumes

1. a) Multiply corresponding components and add: $2 \times 1 + 0 \times 3 + 0 \times 0 = 2$.

2. c) The lengths are 2 and $\sqrt{10}$. Thus $\cos \theta = 2/(2\sqrt{10}) = 1/\sqrt{10}$.

3. d) The first two cross-product components are zero. The third is $2 \times 3 - 0 \times 1 = 6$.

4. c) The area is the magnitude of (0, 0, 6), which is 6. Half of this would be the triangle area.

5. a) The base parallelogram has area 6 and c is perpendicular to it with length 4. Equivalently, the absolute triple product is 24.

6. c) All three vectors now have zero z-component. They lie in the xy-plane, so their scalar triple product and enclosed volume are zero.

## Exercise 2 · Systems of linear equations and algebra

1. a) Subtract $x + y = 14$ from $2x + y = 20$. The y terms cancel and $x = 6$.

2. a) The original equations give $x = 6$ and $y = 8$. Therefore $z = 3 \times 6 - 8 = 10$.

3. b) Both original equations have coefficient 1 on y. Subtracting the complete first equation cancels y without changing the solution set.

4. b) Subtracting the retained first equation gives the new $x = 22 - 14 = 8$. The original x was 6, so the increase is 2.

5. c) Multiplication by 2 must apply to every term on both sides. It gives $2x + 2y = 28$.

6. c) Doubling the first equation gives $2x + 2y = 28$, which contradicts 30. No pair can satisfy both constraints.

## Exercise 3 · Functions and graphs

1. b) The linear function f(x) = 2x + 1 increases by 2 for every unit of x. The other functions do not have a constant slope.

2. c) The squared term is nonnegative and becomes zero at x = 2. This is the vertex and minimum.

3. c) The exponential rule gives $h(3) = 2^3 = 8$.

4. c) Since $2^3 = 8$, $\log_2(8) = 3$.

5. b) Adding 3 to the entire output gives f(x) + 3 = 2x + 4. Changing the input or slope would describe a different transformation.

6. b) At x = 0, both outputs are 1. The other listed pairs are (3,2), (5,4) and (7,8), none of which are equal.

## Exercise 4 · Differentiation and integration

1. d) Differentiate to obtain v(t) = 2t + 2. At t = 2, this is 6 m/s.

2. c) The initial position is 5 and the final position is 20. Their difference, or the integral of velocity, is 15 m.

3. a) Average velocity equals displacement divided by time: 15/3 = 5 m/s. The initial position is not part of the displacement.

4. a) Changing the constant shifts the position reference. Its derivative is zero, so v(t) remains 2t + 2.

5. b) Velocity stays positive on $0 \le t \le 3$. Therefore its integral gives both displacement and distance, unlike an integral of position.

6. c) Differentiate v(t) = 2t + 2 once more. The acceleration is 2 m/s² throughout the interval.

## Exercise 5 · Probability and descriptive statistics

1. c) The total is 2 + 4 + 4 + 6 = 16. Dividing by four outcomes gives a mean of 4.

2. c) The squared deviations from 4 are 4, 0, 0 and 4. Their mean is 2, so the standard deviation is √2.

3. c) The interval is 10 ± 2, exactly one standard deviation on either side of the mean. The passage assigns approximately 68% to this interval.

4. b) Multiply the independent probabilities: 0.2 × 0.2 = 0.04.

5. d) Every reading and the mean increase by 3. Each reading therefore has the same deviation as before, leaving the variance unchanged.

6. c) Three of the four listed outcomes qualify: 4, 4 and 6. Repeated values still represent separate equally likely outcomes.

## Exercise 6 · Matrices and determinants

1. d) The two rows give 2 × 3 + 2 = 8 and 3 + 2 = 5.

2. a) For M, ad - bc = 2 × 1 - 1 × 1 = 1.

3. a) The determinant is 1. Swap the diagonal entries and negate the off-diagonal entries to obtain [[1,-1],[-1,2]].

4. d) Its determinant is 2 × 2 - 4 × 1 = 0. This establishes noninvertibility but does not mean every output is zero.

5. c) Doubling one row doubles the determinant. The original determinant is 1, so the new value is 2.

6. a) Multiplication on the right by N scales the second column of M. Multiplication on the left scales its second row. Thus the order matters.

## Exercise 7 · Kinematics and Newton’s laws

1. a) The net horizontal force is 8 - 2 = 6 N. Dividing by 2 kg gives 3 m/s².

2. d) Starting from rest, v = 0 + 3 × 4 = 12 m/s.

3. d) Use s = ½at² because u = 0: ½ × 3 × 4² = 24 m.

4. a) A third-law pair acts on two different objects. Only forces acting on the trolley enter its own net-force calculation.

5. b) Removing both horizontal forces makes the net force zero. Acceleration is then zero, but the velocity already acquired remains.

6. c) The net force remains 6 N. With double the mass, acceleration becomes 6/4 = 1.5 m/s².

## Exercise 8 · Hydrostatics, pressure and buoyancy

1. c) Add surface pressure to the water contribution: 100,000 + 1,000 × 10 × 5 = 150,000 Pa.

2. c) Gauge pressure excludes surface pressure. Thus it is ρgh = 50,000 Pa.

3. c) Floating requires Vsub = m/ρ = 12/1,000 = 0.012 m³. Divide by total volume 0.020 m³ to obtain 0.60.

4. b) Full submersion displaces 0.020 m³, giving ρgV = 1,000 × 10 × 0.020 = 200 N. This is greater than the block’s weight.

5. a) Constant pV gives V2 = p1V1/p2 = 1 × 2/2 = 1 L. Both pressures are absolute.

6. c) Equilibrium requires Vsub = 12/800 = 0.015 m³. This is below the block’s total volume, so floating is possible.

## Exercise 9 · Heat, temperature and the ideal gas law

1. d) Rearrange $Q = mc\Delta T$: $\Delta T = 80{,}000/(2 \times 4{,}000) = 10$ K.

2. b) At fixed V and n, pressure is proportional to absolute temperature. The temperature ratio is 600/300 = 2.

3. a) With the stated sign convention, $\Delta U = Q - W = 500 - 200 = 300$ J.

4. c) Insulation gives Q = 0, not $\Delta U = 0$. Hence $\Delta U = 0 - 100 = -100$ J.

5. d) Convert the Celsius temperature to an absolute temperature: 27 + 273 = 300 K.

6. d) At fixed pressure and gas amount, V/T is constant. Thus V2 = 2 × 450/300 = 3 L.

## Exercise 10 · Electrical quantities, Ohm’s law and power

1. b) Ohm’s law gives $I = V/R = 12/6 = 2$ A.

2. c) Use $P = V^2/R = 12^2/6 = 24$ W, equivalently 12 V × 2 A.

3. b) At a constant 24 W, $E = Pt = 24 \times 5 = 120$ J.

4. d) With R fixed, power is proportional to $V^2$. Doubling V multiplies power by $2^2 = 4$.

5. c) The total series resistance is 2 + 4 = 6 Ω. The circuit current is 12/6 = 2 A and passes through both resistors.

6. d) The energy is 24 W × 2 h = 48 Wh. Divide by 1,000 to obtain 0.048 kWh.

## Exercise 11 · Stoichiometry, concentration and pH

1. a) The 2 mol $O_2$ consume only 4 mol $H_2$. Hydrogen is available in excess, so oxygen is exhausted first.

2. d) The ratio is 2 mol water per 1 mol oxygen. The limiting 2 mol oxygen therefore produce 4 mol water.

3. c) Use c = n/V = 0.50/2.0 = 0.25 mol/L.

4. c) The solute amount is 0.50 × 0.20 = 0.10 mol. Dividing by the final 1.0 L gives 0.10 mol/L.

5. d) The base-10 logarithm is -3, so its negative is 3.

6. d) The concentration increases tenfold. Applying the logarithmic definition gives pH 2 instead of 3.

## Exercise 12 · Waves and mechanical energy

1. b) Multiply frequency by wavelength: v = 5 × 2 = 10 m/s.

2. a) Ek = ½ × 2 × $3^2$ = 9 J. Speed must be squared.

3. a) Ep = mgh = 2 × 10 × 5 = 100 J.

4. d) Set mgh = ½$mv^2$. Mass cancels, giving v = √(2gh) = √100 = 10 m/s.

5. a) The stated wave model makes energy proportional to amplitude squared. Doubling amplitude therefore gives a factor of four.

6. b) Rearrange v = fλ as λ = v/f. With v fixed, doubling f halves λ.

## Exercise 13 · Statics: forces, moments and equilibrium

1. b) Vertical equilibrium requires the upward support reactions to balance both downward loads: 60 + 40 = 100 N.

2. a) Moments about L give RR = (60 × 1 + 40 × 3)/4 = 45 N.

3. c) The added moment about L is 60 × 1 = 60 N·m. Divide by the 4 m span to obtain a 15 N increase in RR.

4. b) Moment equals force times perpendicular lever arm: 10 × 2 = 20 N·m.

5. a) A couple can have zero net force but a nonzero moment. Both force and moment balance are required for static equilibrium.

6. d) Use the magnitude of one force times the separation: 20 × 0.5 = 10 N·m. Adding the force magnitudes would double-count the couple.

## Exercise 14 · Stress, strain and basic beam behaviour

1. a) Using N and $mm^2$ gives σ = 2,000/20 = 100 $N/mm^2$ = 100 MPa.

2. a) Use the same length units: ε = 0.50/1,000 = 0.0005.

3. a) E = σ/ε = 100/0.0005 = 200,000 MPa, or 200 GPa.

4. a) Doubling area halves stress. With E fixed, strain also halves; since original length is fixed, elongation halves.

5. d) Deflection is proportional to $L^3$. Doubling length gives a factor of $2^3 = 8$.

6. d) The passage explicitly limits this relation to linear elastic behaviour. Units can be converted, but that cannot restore a failed physical assumption.

## Exercise 15 · Series, parallel and Kirchhoff’s laws

1. a) $1/R_p = 1/6 + 1/3 = 1/2$, so $R_p = 2\ \Omega$.

2. b) Total resistance is $R1 + R_p = 2 + 2 = 4\ \Omega$. The source current is $12/4 = 3$ A.

3. a) R1 drops $3\ \text{A} \times 2\ \Omega = 6$ V. The remaining 6 V is across both parallel branches.

4. a) The branch voltage is 6 V. Thus $I3 = 6/3 = 2$ A. The other branch carries 1 A.

5. c) R1 and R2 remain in series with total resistance $8\ \Omega$. Current is $12/8 = 1.5$ A.

6. a) The 3 A entering the junction divides into branch currents of 1 A and 2 A. The voltage equation instead expresses loop voltage balance.

## Exercise 16 · Energy conversion and efficiency

1. d) Motor efficiency is $800/1{,}000 = 0.80$, or 80%.

2. a) The transmission receives 800 J and delivers 600 J. Its efficiency is $600/800 = 75\%$.

3. c) Either divide 600 by 1,000 or multiply 0.80 by 0.75. Both give 60%. Efficiencies of sequential stages are not added.

4. b) The total loss from the useful output is $1{,}000 - 600 = 400$ J, including losses from both stages.

5. d) The motor still delivers 800 J. The improved transmission delivers $0.90 \times 800 = 720$ J.

6. d) For steady power, efficiency is useful output divided by input: $300/500 = 60\%$.

## Exercise 17 · Control and feedback loops

1. a) Subtract the measurement from the target: e = 60 - 52 = 8 °C.

2. b) The computed command is 2 × 8 = 16, which lies within the actuator limits and is not clipped.

3. c) The new error is 3 °C, so the command is 2 × 3 = 6 units.

4. c) The report equals actual temperature plus 5. Therefore actual temperature is 60 - 5 = 55 °C. This does not assume steady-state error is zero.

5. b) Open-loop operation omits measurement-based adjustment. Saturation or a different gain does not itself remove the feedback loop.

6. a) The computed command is 30, but the actuator clips it at the maximum of 20.

## Exercise 18 · Unit conversions and dimensional analysis

1. a) Convert distance and time: 72 × 1,000/3,600 = 20 m/s.

2. c) One g/cm³ is 0.001 kg divided by 0.000001 m³, or 1,000 kg/m³. Multiply by 2.

3. b) Square the length conversion before multiplying: 3 × $(0.01)^2$ = 0.0003 m².

4. a) Convert 2 cm² to 0.0002 m². Then pressure = 100/0.0002 = 500,000 Pa.

5. d) The units of $v^2/g$ are $(m^2/s^2)/(m/s^2)$ = m. In contrast, v/g has units s, vg has units $m^2/s^3$, and $g/v^2$ has units 1/m.

6. a) Multiply 0.5 by the stated 3.6 × $10^6$ J per kWh to obtain 1.8 × $10^6$ J.

## Exercise 19 · Binary and hexadecimal number systems

1. b) The occupied places are 8, 4 and 1. Their sum is 13.

2. d) 26 = 16 + 8 + 2, so the binary digits are 11010.

3. d) The first digit contributes 2 × 16 and F contributes 15. The result is 47.

4. d) Pad to 0010 1000. The groups correspond to hexadecimal digits 2 and 8.

5. a) 1111 represents 15. Adding 1 gives 16, but modulo-16 storage discards the carry and stores 0.

6. b) Move each bit one position left and insert a zero on the right: 0011 becomes 0110.

## Exercise 20 · Boolean logic and logic gates

1. c) B OR C is 1 because C is 1. Then A AND 1 is also 1.

2. b) An AND gate with one input equal to 0 produces 0. Therefore disabling A forces the indicator output to zero.

3. a) The first identity given in the passage is De Morgan’s relation. Both B and C must be zero for the negated OR to be one.

4. a) The inputs are equal, so XOR produces 0. OR would produce 1 for this same pair.

5. b) The distributive identity gives (A AND B) OR (A AND C): OR the outputs of the two AND gates. The other circuits differ on some input combinations.

6. a) A must be 1. Of the four B,C pairs, 01, 10 and 11 make B OR C equal 1. Thus exactly three combinations qualify.

## Exercise 21 · Algorithms and step-by-step tracing

1. b) The array uses indices 0, 1, 2 and 3. The value 9 is at index 2.

2. d) The target is absent. The search checks all four elements before returning -1.

3. b) First swap 7 and 2. Next compare 7 and 9 without swapping. Finally swap 9 and 4, producing [2,7,4,9].

4. a) The decisions to discard a half rely on ascending order. The other listed properties are not required by the described routine.

5. a) The initial bounds are 0 and 3, so the middle index is floor(3/2) = 1. Its value is 4.

6. d) The routine stops at the first match. The first element has value 4 and index 0.

## Exercise 22 · Algorithmic complexity and growth

1. b) In $3n + 5$, the linear term dominates for large n. Constant multipliers and additive constants do not change the tightest listed growth class.

2. d) The $n^2$ term dominates $n^2 + n$. Replacing n by $2n$ makes that dominant term four times as large.

3. c) $\log_2 8 = 3$, which is already an integer. Adding 1 gives four operations in this model.

4. b) For each of n outer iterations, the inner loop performs n operations. The total is $n \times n = n^2$.

5. d) Sequential work adds rather than multiplies. The n log n sorting term dominates the additional logarithmic search.

6. a) $TA(1) = 8$ and $TB(1) = 2$. A better asymptotic growth class does not guarantee a lower count for every small input.

## Exercise 23 · Arrays, stacks and queues

1. b) B was pushed last and is at the top. The stack removes and returns B.

2. d) A entered first and is at the front. The queue removes and returns A.

3. a) Zero-based indices 0, 1, 2 and 3 correspond to values 9, 5, 7 and 1.

4. c) The first pop removes 5, leaving 3. Pushing 7 puts it on top, so the final pop returns 7.

5. d) A queue preserves arrival order for removal. A stack reverses the order of the most recently stored actions.

6. c) The first dequeue removes X, leaving the queue empty. The specified model reports underflow on another removal.

## Exercise 24 · Recursion and iteration

1. d) Expand to 3 + 2 + 1 + F(0) = 3 + 2 + 1 + 1 = 7.

2. a) The nonnegative integer input decreases on each recursive step and reaches the explicitly defined base case.

3. c) Starting with 1, the procedure adds 1, then 2, then 3, producing 7.

4. c) The calls are F(4), F(3), F(2), F(1) and F(0), giving five in total.

5. b) G(5) = 2 + G(3) = 2 + 2 + G(1) = 5.

6. c) Only i = 1 and i = 2 are included. Starting from 1, the final value is 1 + 1 + 2 = 4.

## Exercise 25 · Optimal order quantity and inventory costs

1. d) $Q^* = \sqrt{(2 \times 3{,}600 \times 25/2)} = \sqrt{90{,}000} = 300$ units.

2. b) Stock decreases steadily from 300 to zero. Its average over a replenishment cycle is $Q/2 = 150$ units.

3. c) Annual orders equal $D/Q = 3{,}600/300 = 12$.

4. a) S is inside the square root. Multiplying S by 2 multiplies $Q^*$ by $\sqrt{2}$ rather than by 2.

5. a) Ordering cost is $3{,}600 \times 25/300 = €300$. Holding cost is $2 \times 300/2 = €300$. Their sum is €600.

6. c) A quantity discount makes purchase cost depend on order size. That conflicts with an explicit assumption and changes the objective that must be compared.

## Exercise 26 · Break-even analysis

1. d) Subtract unit variable cost from price: 50 - 30 = €20.

2. b) Divide fixed cost by unit contribution: 12,000/20 = 600 units.

3. d) The total contribution is $20 \times 900 = €18{,}000$. Subtract €12,000 fixed cost to obtain €6,000 profit.

4. c) The new unit contribution is 55 - 30 = €25. Break-even quantity is 12,000/25 = 480.

5. d) The unit contribution falls to 50 - 35 = €15. Break-even quantity becomes 12,000/15 = 800.

6. a) Maximum contribution is $500 \times €20 = €10{,}000$, below €12,000 fixed cost. The resulting operating profit is -€2,000.

## Exercise 27 · Accounting statements and transaction logic

1. a) Initial assets are €30,000. Subtract liabilities of €12,000 to obtain equity of €18,000.

2. b) Profit uses earned revenue, not only collected cash: €8,000 - €5,000 = €3,000.

3. a) The business earned €8,000 but collected €5,000, leaving €3,000 due from customers.

4. b) Cash is an asset and the repayment obligation is a liability. Borrowing does not itself earn revenue or create profit.

5. c) One asset increases by exactly the amount another decreases. Under the stated model, the purchase is not immediately expensed.

6. a) Depreciation reduces the equipment’s carrying value and period profit. With no other transactions, the expense also reduces equity; it is not a current cash payment.

## Exercise 28 · Marketing mix, pricing and campaign comparisons

1. c) Durability is a feature of the item itself, so it is primarily a product decision.

2. d) Distribution and customer access fall under place. This differs from communicating an offer through advertising.

3. c) A produces 1,000 × 2% = 20 orders. Contribution is (20 - 8) × 20 - 150 = €90.

4. c) B produces 30 orders. Its contribution is (16 - 8) × 30 - 150 = €90, the same as A.

5. b) Before advertising, the campaign contributes (20 - 8) × 20 = €240. Spending exactly this amount on advertising leaves zero contribution.

6. b) Week and audience also changed. The arithmetic comparison is valid, but it does not isolate a causal price effect.

## Exercise 29 · Project scheduling and the critical path

1. d) A finishes at 3, B at 7 and C at 5. D finishes at 12 and E at 7. F starts at 12 and finishes at 13.

2. a) A-B-D-F takes 3 + 4 + 5 + 1 = 13 days. The other valid listed paths are shorter; B is not a predecessor of E.

3. a) C then finishes at 6. B still controls D's start at 7, and E finishes by 8, before D. F still finishes at 13.

4. a) B finishes at 9, delaying D to finish at 14. F then finishes at 15.

5. c) D requires both B and C. Their finish times are 7 and 5, so D must wait until day 7.

6. d) E was already finished before the critical D branch. Shortening E does not change F's controlling predecessor or the 13-day duration.

## Exercise 30 · Process capacity, bottlenecks and flow

1. d) The smallest stage capacity is assembly's 40 units/hour. Every unit must pass through it.

2. b) The capacities become 60, 55 and 50. Inspection becomes the bottleneck at 50 units/hour.

3. a) Demand is below the 40-unit line capacity. Actual sustained sales-supported output is therefore limited to 35 units/hour.

4. a) Assembly processes 35 units/hour out of its capacity of 40: 35/40 = 87.5%.

5. a) Convert 30 minutes to 0.5 hour. Little's law gives L = 40 × 0.5 = 20 units.

6. d) More waiting stock does not change assembly's 40-unit/hour processing limit. Each listed capacity increase would raise the original line limit, up to inspection's 50.

## Exercise 31 · Supply, demand and market equilibrium

1. d) Set 100 - 2P = 20 + 2P. This gives 80 = 4P and P = 20.

2. d) At P = 20, demand is 100 - 40 = 60 and supply is 20 + 40 = 60.

3. b) Demand is 70 and supply is 50, giving a shortage of 20 units.

4. c) Set 120 - 2P = 20 + 2P. Then 100 = 4P and P = 25.

5. d) The new price is (100 - 40)/4 = 15. Quantity demanded at that price is 100 - 30 = 70.

6. c) At P = 10, Qd = 80 and Qs = 40. The shortage is 80 - 40 = 40 units.

## Exercise 32 · Price elasticity and revenue

1. d) The quantity-change magnitude is 20/90. The price-change magnitude is 2/11. Their ratio is 11/9, approximately 1.22.

2. b) The midpoint elasticity is approximately 1.22, which exceeds 1.

3. b) Initial revenue is €1,000 and final revenue is €960. The change is -€40.

4. c) Use the local approximation: -0.5 × 10% = -5%.

5. b) With zero quantity response, elasticity is zero. On the stated axes, the demand curve is vertical.

6. d) At E = 1, the percentage quantity decrease approximately offsets the percentage price increase. This is a first-order statement, not an exact finite-change identity.

## Exercise 33 · Cost curves and profit maximisation

1. d) Set 2 + 2q = 12, giving q = 5. Increasing marginal cost makes this the profit-maximising interior output.

2. b) TC(5) = 20 + 2 × 5 + $5^2$ = €55.

3. b) Revenue is 12 × 5 = €60. Subtract total cost €55 to obtain profit €5.

4. a) The added fixed cost has zero derivative, so MC is unchanged. The output solving MC = MR remains 5 even though profit falls.

5. d) VC(5) = 10 + 25 = €35. Divide by five units to obtain €7 per unit.

6. d) Profit becomes 2q - (20 + 2q + $q^2$) = -20 - $q^2$. This is largest at q = 0.

## Exercise 34 · GDP, inflation and unemployment measures

1. d) Use year-2 quantity and year-2 price: 110 × €12 = €1,320.

2. b) Real GDP at the base price is 110 × €10 = €1,100.

3. d) Real GDP rises from €1,000 to €1,100. Growth is 100/1,000 = 10%.

4. c) The price index rises from 100 to 120, so inflation is 20%. This differs from nominal GDP growth.

5. d) The labour force is 920 + 80 = 1,000. The unemployment rate is 80/1,000 = 8%.

6. b) Unemployment becomes 60 and the labour force becomes 980. The rate falls to about 6.12%, while employment remains 920.

## Exercise 35 · Market structures and monopoly decisions

1. d) The passage explicitly identifies firms under perfect competition as taking market price as given.

2. c) Price is 20 - q and marginal revenue is 20 - 2q. Their difference is q, so MR is lower whenever q > 0.

3. b) Set 20 - 2q = 4, giving 2q = 16 and q = 8.

4. d) Insert q = 8 into inverse demand: P = 20 - 8 = €12. The firm does not charge marginal revenue as its price.

5. c) Set P = 4 in P = 20 - q to obtain q = 16.

6. d) Strategic interdependence among a few important sellers is the defining feature of oligopoly in the passage.

## Exercise 36 · Game theory and payoff matrices

1. b) Given Column’s C, Row receives 3 from C and 5 from D. Thus D is the best response.

2. a) Against C, D gives 5 rather than 3. Against D, D gives 1 rather than 0. D is better in both cases.

3. c) Both organisations prefer D against either opponent choice. At (D,D), switching alone to C lowers own payoff from 1 to 0.

4. d) The joint payoff is 6 at (C,C), 5 at either mixed-choice outcome, and 2 at (D,D). Mutual cooperation has the largest total, despite not being individually stable.

5. c) With Column still choosing D, the relevant cell becomes (C,D), whose Row payoff is 0.

6. b) Against C, defecting still yields 5 rather than 4. Against D, it still yields 1 rather than 0. Both organisations retain strictly dominant D.

## Exercise 37 · Qualitative and quantitative research strategies

1. d) It starts with a hypothesis and tests a numerical relationship using standardised data, matching the stated quantitative, deductive approach.

2. a) Detailed accounts can reveal processes and conditions. Twelve deliberately selected interviews do not support the other population-level or experimental claims.

3. b) The passage allows an iterative qualitative process when changes are made transparent. Refinement does not itself establish prevalence or invalidate all prior data.

4. c) The issue is concealing the timing of hypothesis formation. Exploratory work can be useful, but should not be presented as a pre-specified confirmation.

5. b) Numerical records address patterns across cases, while interviews explore how those patterns could arise. The combination directly fits the two research aims.

6. b) Participation was not random. Selection and other differences may contribute to the association, so a causal effect is not established by this finding alone.

## Exercise 38 · Sampling and statistical significance

1. b) The population is divided into strata, and random samples are selected within them. Equal sample sizes need not match population shares.

2. c) Weight by the population composition: $0.80 \times 0.40 + 0.20 \times 0.80 = 0.48$, or 48%.

3. b) The p-value describes a probability of data at least as extreme conditional on the null model. It is not a posterior probability of either hypothesis.

4. a) Because $0.03 < 0.05$, the pre-specified rule rejects the null. This decision does not establish certainty or justify changing the threshold.

5. a) The null asserts equal population rates. Rejecting it when that assertion is true is a Type I error.

6. c) A larger sample from the same selective source can retain systematic bias. Sample size alone does not repair an unrepresentative recruitment process.

## Exercise 39 · Correlation, confounding and causal claims

1. b) A positive correlation describes an average direction of linear association. It does not imply universal ordering or a causal dose effect.

2. c) A common cause can create or contribute to an observed association between two other variables.

3. b) This explanation runs from an academic outcome or experience toward attendance, rather than assuming attendance is always the cause of performance.

4. b) Randomisation concerns the offer, so the clean comparison is between assignment to the offer and assignment to no offer.

5. a) Variables measured at the same time may leave temporal order unresolved, even when their association can be calculated.

6. b) The passage limits r to linear association. A zero value does not rule out a curved relationship or establish complete independence.

## Exercise 40 · Survey and questionnaire design

1. c) A respondent may find the service reliable but expensive, or unreliable but cheap. One answer cannot clearly separate the two constructs.

2. c) “How would you rate the service’s punctuality?” asks for an assessment without supplying a positive or negative evaluation in advance.

3. c) The categories 0; 1-2; 3-4; 5 or more include every nonnegative whole-number count exactly once. The other sets either overlap, omit zero or omit counts above four.

4. a) A common explicit interval makes the frequency question more consistently interpretable across respondents.

5. a) Measurement changed along with time. An observed response difference may therefore mix a real trend with a wording effect.

6. d) Random assignment with other procedures held constant provides the clearest comparison of wording itself among the listed designs.

## Exercise 41 · Argument analysis and logical validity

1. b) The premise says every approved design has passed this load test. The other conclusions go beyond that premise.

2. a) Passing is necessary but not stated to be sufficient. K could still fail another approval requirement.

3. d) If L were approved, it would have passed. Not passing therefore implies not approved, using the valid contrapositive pattern.

4. d) A counterexample to “passed implies approved” must satisfy passed while failing approved.

5. c) The reviewer substitutes a personal attack for examination of the estimate and its supporting reasons.

6. b) The criticism attacks a distorted, more extreme claim. The original proposal concerns unnecessary printing, not a ban on communication.

## Exercise 42 · Social and psychological concepts in applied research

1. b) The message describes what others commonly do. It does not explicitly state what they approve of or provide a reward.

2. d) The message group has 300 returns among 500 students, giving 60%.

3. b) The rates are 60% and 40%. Their difference is 20 percentage points; the relative increase is a different quantity.

4. d) The target behaviour is returning a cup. An observed return count measures it directly, whereas the other measures concern attitudes or unrelated recognition.

5. c) Random assignment and the stated controls support a causal comparison of the message here. They do not establish universal generalisation or an exact mechanism.

6. b) A desirable consequence contingent on a behaviour that increases its future occurrence fits the stated definition of positive reinforcement.